

cPanel Deployment Guide

Comprehensive guide for hosting Vite (React) + Node.js (Express + Prisma + MySQL) on cPanel

1. Architecture & Domains Checklist

To host your decoupled application cleanly, configure your domains as follows:

- **Frontend URL:** `https://yourdomain.com` (Root Directory: **public_html**)
- **Backend API URL:** `https://api.yourdomain.com` (Root Directory: **crystal-api** or **public_html/api**)

2. Step 1: Set Up MySQL Database in cPanel

1. Log in to cPanel and search for **MySQL Database Wizard** under Databases.
2. **Create a Database:** Enter a name (e.g., *crystal_db*) and click Next.
3. **Create User:** Enter a username (e.g., *crystal_user*), generate a strong password, and save these credentials.
4. **Privileges:** Check **ALL PRIVILEGES** to associate the user with the database, and click Make Changes.
5. Go back to cPanel Home, open **phpMyAdmin**, click on your new database, select the **Import** tab.
6. Choose your local file **crystal_backup.sql** and click **Import**.

3. Step 2: Build & Upload Vite React Frontend

A. Configure Frontend Environment:

Create a `.env` file in the root of your frontend folder locally with the production API URL:

```
VITE_API_BASE_URL=https://api.yourdomain.com
```

B. Build the Project:

Run the build command locally:

```
npm run build
```

C. Upload Static Assets:

1. Open cPanel **File Manager** and navigate to **public_html**.
2. Compress the contents of your local **dist/** folder into a ZIP file.
3. Upload and extract it inside **public_html**.

D. Configure Client-Side Routing (.htaccess):

Create a file named **.htaccess** in **public_html** and add the following rule to handle React router reloads:

```
<IfModule mod_rewrite.c>
RewriteEngine On
RewriteBase /
```

```
RewriteRule ^index\.html$ - [L]
RewriteCond %{REQUEST_FILENAME} !-f
RewriteCond %{REQUEST_FILENAME} !-d
RewriteRule . /index.html [L]
</IfModule>
```

4. Step 3: Compile and Upload the Backend

A. Compile TypeScript:

In your local server folder, compile TypeScript to JavaScript:

```
npm run build
```

B. Select files to upload:

Upload only the following files/directories to the backend root (e.g., *crystal-api*):

- **dist/** (Compiled JS code)
- **prisma/** (Database schema)
- **package.json** and **package-lock.json**

5. Step 4: Host the Node.js Backend on cPanel

1. Create a subdomain in cPanel (e.g., **api.yourdomain.com**) and point its document root to a folder named **crystal-api**.
2. Go to cPanel and open **Setup Node.js App**.
3. Click **Create Application** and enter the following settings:
 - **Node.js Version:** Select 18.x or 20.x
 - **Application Mode:** Production
 - **Application root:** crystal-api
 - **Application URL:** api.yourdomain.com
 - **Application startup file:** dist/index.js
4. Scroll to **Environment variables** and add:
 - **DATABASE_URL:** mysql://username:password@localhost:3306/database_name
 - **NODE_ENV:** production
 - **JWT_ACCESS_SECRET & JWT_REFRESH_SECRET:** (secure random keys)
 - **ALLOWED_ORIGINS:** https://yourdomain.com,https://www.yourdomain.com
5. Click ****Create****.
6. Upload the backend files to the *crystal-api* directory.
7. Go back to **Setup Node.js App**, click Edit on your application, and click **Run JS Install** to install packages.
8. In cPanel Terminal, run **npm prisma generate** in the *crystal-api* directory to generate the Prisma client.
9. Click **Restart** at the top of the Setup Node.js App page.

6. Step 5: Troubleshooting Tips

- **CORS Errors:** Ensure your **ALLOWED_ORIGINS** exactly matches your frontend domain URL.
- **Database Connection Error:** Verify your database username, database name, and password match exactly in the **DATABASE_URL** environment variable.
- **404 On Page Refresh:** Make sure the **.htaccess** file exists in your frontend's **public_html** folder.

- **Node.js Error logs:** View the stderr.log inside the crystal-api directory in cPanel File Manager.